Sequences and Basic Series Convergence Tests

Section 1: Limits of Sequences & Monotonicity

1 Determine the limit of the sequence $a_n = \frac{4n^2 - 3n}{2n^2 + 5}$.

a. 2

b. 4

c. ∞ (diverges)

d. 0

2 Which statement is guaranteed by the Monotone Convergence Theorem for Sequences?

a. Every bounded monotonic sequence converges

b. Every alternating sequence converges

c. Every bounded sequence is monotonic

d. Every convergent sequence is positive

3 What is $\lim_{n \to \infty} (1 + \frac{2}{n})^n$?

a. e^2

b. e

c. 1

d. ∞

Section 2: Geometric Series & n-th Term Divergence Test

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4 Evaluate the sum of the series $\sum_{n=1}^{\infty} 3(\frac{2}{5})^n$.

a. 2

b. $\frac{6}{5}$

c. $\frac{15}{2}$

d. $\frac{5}{2}$

5 If $\lim_{n\to\infty} a_n = \frac{1}{2}$, what does the n-th Term Test conclude about $\sum_{n=1}^{\infty} a_n$?

a. The series diverges

b. The series converges to $\frac{1}{2}$

c. The test is inconclusive

d. The series is geometric

6 If $\lim_{n\to\infty} a_n = 0$, what can be concluded about $\sum a_n$?

a. The test is inconclusive; the series may converge or diverge

b. The series converges

c. The series diverges

d. The sum is zero

Section 3: Integral Test & p-Series

7 For what values of p does the p-series $\sum_{n=1}^{\infty} \frac{1}{n^p}$ converge?

a. $p > 1$

b. $p \geq 1$

c. $p < 1$

d. $p > 0$

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8 Which series converges by the Integral Test?

a. $\sum_{n=2}^{\infty} \frac{1}{n(\ln n)^2}$

b. $\sum_{n=2}^{\infty} \frac{1}{n \ln n}$

c. $\sum_{n=1}^{\infty} \frac{1}{\sqrt{n}}$

d. $\sum_{n=1}^{\infty} \frac{1}{2n+1}$

9 What three conditions must $f(x)$ satisfy on $[1, \infty)$ to apply the Integral Test to $\sum a_n$?

a. $f(x)$ must be continuous, positive, and decreasing

b. $f(x)$ must be differentiable, alternating, and bounded

c. $f(x)$ must be positive and increasing

d. $f(x)$ must be a rational function

Section 4: Direct Comparison & Limit Comparison Tests

10 Determine the convergence of $\sum_{n=1}^{\infty} \frac{1}{n^3+5n}$.

a. Converges by Direct Comparison with $\sum \frac{1}{n^3}$

b. Diverges by Comparison with $\sum \frac{1}{n}$

c. Diverges by n-th term test

d. Inconclusive

11 Which series should be used for the Limit Comparison Test with $\sum_{n=1}^{\infty} \frac{3n+2}{2n^3-1}$?

a. $\sum \frac{1}{n^2}$ (convergent)

b. $\sum \frac{1}{n}$ (divergent)

c. $\sum \frac{1}{n^3}$ (convergent)

d. $\sum n$ (divergent)

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12 In the Limit Comparison Test, if $\lim_{n\to\infty} \frac{a_n}{b_n} = L$ where $0 < L < \infty$, what does this mean?

a. Both $\sum a_n$ and $\sum b_n$ either both converge or both diverge

b. $\sum a_n$ converges to L

c. $\sum a_n$ converges while $\sum b_n$ diverges

d. The test is inconclusive

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